

WILLIAM (BILL) XIA

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QUALIFICATIONS

- **4+ years of experience** with full-stack AI software development, natural language processing (NLP), large language models (LLMs), data processing, and human-computer interaction (HCI).
- **First-author of 1 paper presented at ACL 2025** and 1 thesis paper; co-author of 3 additional papers.

EDUCATION

Bachelor of Science in Computer Science, Tufts University Sept 2021 – May 2025

Minors in Mathematics & English

GPA: 3.81 / 4

Honors: *Summa cum laude*; Senior Honors Thesis; Dean's List (all semesters)

Teaching: Programming Languages TA (4 semesters); led weekly recitations and office hours for 30+ students.

WORK EXPERIENCE

Maven Bio

Boston, MA

Founding Applied AI Engineer | *Claude Agent SDK • MCP • Multi-Agent Systems* Oct 2025 – Present

- **Built Maven's flagship research agent** (Claude Agent SDK in sandboxed E2B env), contributing sub-agents, MCP tools, and agent skills for building competitive landscapes and differentiating assets; a client reported that this agent reduced human work by over 9 weeks per FTE annually.
- **Led development of Maven Monitors**, an autonomous, multi-agent system that maintains living landscapes of drugs, trials, and companies; designed and shipped end-to-end, including Celery scheduler and frontend UI. Now runs over 200 living landscapes, surfacing ~110 news signals per day.
- **Owned Maven's agentic data-QA fleet** to review new and standing records across 11 data surfaces; autonomous QA processes ~1,400 records per day, requiring human intervention for only 1.7% of cases.

Dassault Systèmes, SolidWorks Product Management Team

Waltham, MA

AI Product Management Intern | *Local LLMs • Tool-Use Agents • Full-Stack Dev.* June 2025 – Sept 2025

- **Developed a full-stack, local LLM agent** that interfaces with Dassault's internal web APIs; the agent improved access to analytics and documentation for 20+ SolidWorks project managers.
- **Deployed the AI assistant** within the Jenkins CI/CD pipeline, enabling secure, scalable access across internal cloud systems and reducing manual data retrieval time by 25%.

National Library of Medicine, Natural Language Processing Lab

Bethesda, MD

Research Intern | *Model Fine-Tuning • Dataset Design • LLM Benchmarking* June 2023 – May 2025

- **Designed and released JEBS**, a novel biomedical lexical simplification task and dataset comprising 21,595 annotated term-replacement pairs derived from 400 biomedical abstracts; developed end-to-end simplification system using BERT, Llama, and GPT-4o. Advised by Dr. Dina Demner-Fushman.
- **First-authored a paper** on JEBS presented at ACL 2025; co-authored a companion paper submitted to Journal of Biomedical Informatics and presented its findings at the 2024 Text Retrieval Conference.
- **Led development** of JEBS baseline models, which achieved strong domain-expert human evaluation scores across simplicity, accuracy, and brevity.

FEATURED PROJECTS

Stone Brain | *Knowledge Graphs • Relation Extraction • LLM Evals*

Aug 2026

- **Built a document-consistency manager** that uses an LLM to extract structured claims and entity relationships from a text corpus, then uses them to construct a Postgres knowledge graph.
- **Designed a conflict-adjudication system** for detecting and resolving contradictory claims.

Senior Honors Thesis | *Reasoning • World Models • Reinforcement Learning*

Sept 2024 – May 2025

- **Conducted research** under Prof. Vasanth Sarathy and Prof. Matthias Scheutz on grounding LLM agents with natural language world models for embodied reasoning and creative problem solving.
- **First-authored a 57-page thesis paper** compiling my methods and results.

SKILLS

Programming Languages: Python, Java, C/C++/C#, TypeScript, Linux, SQL, OCaml

Other Technical: Claude Agent SDK, MCP, Hugging Face, Ollama, LangChain, Celery, Next.js, React, Git

Spoken Languages: English, Mandarin